Selahaddin Islamoglu

Software Developer

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Summary

8 years of expertise in developing communication applications, software, and firmware solutions. Demonstrated proficiency in collaborating effectively with diverse international teams. Highly skilled in C programming for Linux and RTOS platforms. Experienced in C++ and Go (Golang).

Skills

Languages	C, C++, Go (Golang), Bash, SQL
System	POSIX APIs, multithreading, inter-process communication (IPC), synchronization (mutexes, condition variables), socket programming (TCP/UDP/raw sockets), memory management, I/O
Backend	RESTful APIs, gRPC, GraphQL, JWT, authentication, authorization, database integration (SQL/NoSQL), message queues, telemetry, microservices and serverless architecture.
Protocols	HTTP, FTP, MQTT, SIP, WebRTC, SNMP, LwM2M, OneM2M, NETCONF TCP, UDP, SSL, RTP, RTCP, ICMP, STUN, TURN
Platforms	GNU/Linux, FreeRTOS, Zephyr OS, AWS (S3, EventBridge, SQS, Lambda), Docker, OpenSIPS
Libraries	glibc, pthread, libcurl, libwebsockets, libevent, OpenSSL, mbedTLS, ncurses Gin, Fiber, Echo, Gorilla Mux, GORM, sqlx, Cobra, Viper, Zerolog, Testify
Tools	Git, Jenkins, Make, CMake, Go Modules, GCC, GDB, Valgrind, Wireshark
Links	Github , HackerRank , Developer Roadmap

Experience

Self-Development & Career Growth

Software Developer — APR 2025 - PRESENT

- Investing dedicated time in structured self-development to strengthen technical skills.
- Building a solid foundation in C++ and Go, focusing on language fundamentals, clean coding practices, and real-world applications.
- Strengthening software development fundamentals, including data structures, algorithms, design patterns, and version control.
- Exploring software engineering concepts such as system design, APIs, and modern backend practices.
- Applying learning through hands-on projects to deepen practical understanding.
- Expanding knowledge of tools, frameworks, and libraries within C++ and Go ecosystems.
- You can track my journey from Developer Roadmap , GitHub , and HackerRank profiles.

Orion Innovation Inc.

Software Developer — JUL 2023 - MAR 2025

Project: Contact Center Automatic Call Distribution

- Worked for Avaya - Contact Center Extended Capacity product on the behalf of Orion Innovation.
- Participated in the takeover of a legacy project from a third-party vendor; contributed to resolving inherited technical issues during the transition.
- Focused on optimizing traffic switchover from active to standby systems during failure scenarios, improving system reliability and performance.
- Actively worked with SIP messaging protocols and OpenSIPS for call routing and signaling.
- Designed and implemented a data persistence capability for the Call Detailed Recording (CDR) feature across active and standby nodes. Developed user-space applications on Linux using C++.

Project: Communication Server

- Worked for Ribbon Communications - C20 Call Controller product on the behalf of Orion Innovation.
- Worked on the platform module that underpins the communication server. Resolved gating bugs to support a successful and timely release.
- Addressing issues related to software management features such as install, upgrade etc. Utilized C++ for development on a Linux environment.

Cavli Wireless Inc.

Firmware Developer — JUL 20 – AUG 21 | COVID DISEASE BREAK | JUN 22 – NOV 22

Project: Cellular IoT Modules

- Worked for development of Cavli Wireless - C-Series IoT Modules product.
- Designed and implemented communication protocol APIs (FTP, HTTP, MQTT, etc.) for AT command interface to enhance system connectivity capabilities.
- Worked on the development of a cloud-based device management system for IoT modules.
- Successfully integrated these features into separate modules with diverse infrastructures.
- Programmed in C for both Linux and Real-Time Operating Systems (RTOS).

Netas Telecommunication Inc.

Software Developer — AUG 15 – DEC 17 | MILITARY SERVICE BREAK | AUG 18 – JUN 20

Project: WebRTC – SIP Gateway

- Worked for Ribbon Communications - Kandy Link product on the behalf of Netas Telecommunication.
- Developed RTP media transportation layer capabilities and performance enhancement.
- Worked with communication protocols such as SIP, WebRTC, STUN, TURN, RTP, and RTCP.
- Implemented media layer modules for user space Linux applications using C programming.
- Gained experience in Java by contributing to the development of signaling modules for the gateway.

Project: Multi Access EDGE Computing Gateway

- An R&D project, developed with the support of TUBITAK.
- Participated in an edge gateway module implementation for the newly announced MEC standard.
- Developed a NETCONF configuration module for efficient configuration management.
- Used C for user space applications on Linux.

Project: OneM2M IoT Gateway

- An R&D project, developed with the support of TUBITAK.
- Participated a gateway module implementation for the newly announced OneM2M standard.
- Developed registration and data storage algorithms for user space applications on Linux using C programming.

Education

Bachelor's degree in Electronics Engineering

Istanbul Technical University

ISTANBUL, TURKEY — 2009 - 2014

Graduate Project: Virtual Instrumentation Based Oscilloscope

The oscilloscope consist of 2 part ; signal processing card and computer interface. The signal data sampled in a time windows on card is displayed according to desired criteria on the computer screen.

Internship Projects: Pool ionization system, Heart rate monitor, TURKSAT 3USAT Cube Satellite Communication Board

Student Projects: Chess playing robot arm, Retro dotmatrix game console, 8x8x8 LED cube.

Voluntering: Head of Laboratory Committee and Executive Board Member at ITU IEEE Student Branch

Projects

Avaya - Contact Center Extended Capacity

Single-server solution for large contact centers supporting 30,000 agents and 60,000 UC users with up to 57,000 calls concurrently. Solution Description

Ribbon Communications - C20 Call Controller

Provides call and session control, network connectivity, switching, signaling, protocol interworking, voice and communications for service providers. Solution Description

Cavli Wireless - C-Series IoT Modules

Cavli C-Series IoT modules are enterprise grade cellular connectivity modules with built-in eSIM/iSIM technology, pre-loaded with global cellular connectivity subscriptions powered by Cavli Hubble.

Solution Description

Ribbon Communications - Kandy Link

Kandy Link is a WebRTC -SIP gateway that uses REST signaling on the web side and maps the REST signaling into compliant SIP signaling on the SIP communications network side. Solution Description